

# AEGIS Supplementary Material: Terminology, Techniques, and Evaluation Metrics

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## Purpose

This supplementary note defines every metric used in the AEGIS evaluation and ablation tables. Metrics are reported at the query, candidate, answer, or system level as indicated below.

## Terminology and techniques

The following entries define the model names, processing tools, retrieval components, and system techniques introduced in the main manuscript and Figure ??; the table is provided as supplementary material because these terms are implementation details rather than new evaluation metrics.

Table 1: Terminology and techniques used in AEGIS.

| Term                     | Category                 | Meaning and role in AEGIS                                                                                                                       |
|--------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| WeMM-Embedding-4B        | Multimodal encoder       | Tencent image-text embedding model producing unified 2,048-dimensional vectors for visual and textual retrieval.                                |
| SigLIP                   | Vision-language encoder  | Complementary visual encoder whose similarity scores provide a second dense retrieval signal.                                                   |
| PP-OCRv6                 | OCR system               | Optical character recognition pipeline that extracts on-screen text and adds it to the searchable payload.                                      |
| Whisper / faster-whisper | Speech recognition       | Automatic speech recognition used to produce searchable transcripts from video audio.                                                           |
| Qdrant HNSW              | Vector index             | Approximate nearest-neighbor index for low-latency dense retrieval; HNSW denotes Hierarchical Navigable Small World graphs.                     |
| BM25                     | Lexical retrieval        | Term-weighted full-text ranking over the enriched text payload, complementing dense similarity.                                                 |
| RRF                      | Rank fusion              | Weighted Reciprocal Rank Fusion combining dense, lexical, and secondary-encoder rankings.                                                       |
| CQR / HyDE               | Query expansion          | Contextual Query Rewriting preserves conversational constraints; Hypothetical Document Embeddings provide an optional expanded retrieval query. |
| MRL                      | Representation technique | Matryoshka Representation Learning supports truncated embedding dimensions for speed-recall trade-offs.                                         |
| Fail-closed VQA          | Safety policy            | Returns UNKNOWN/N/A when media provenance or visual evidence is unavailable, preventing unsupported answers.                                    |

## Reporting conventions

All percentages are reported on the evaluated query corpus and collection snapshot. Latency values include only the stages named in the corresponding experiment. Approximate HNSW results and exact brute-force results are compared using the same query vectors and candidate collection. For VQA, missing or invalid provenance produces the explicit fail-closed response UNKNOWN/N/A with zero confidence rather than an unsupported answer.

Table 2: Evaluation metrics used by AEGIS.

| Metric                     | Level                        | Definition and interpretation                                                                                                                                                 |
|----------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recall@k                   | Query / retrieval            | Fraction of queries whose ground-truth item appears within the first $k$ ranked candidates. We report Recall@1, Recall@5, and Recall@10 where available.                      |
| Mean Reciprocal Rank (MRR) | Query / retrieval            | Mean of the reciprocal rank of the first relevant result: $MRR = N^{-1} \sum_i 1/r_i$ . Higher is better; it rewards earlier correct candidates.                              |
| RRF score                  | Candidate / fusion           | Weighted Reciprocal Rank Fusion score used to combine dense, lexical, and secondary-encoder rankings: $RRF(d) = \sum_m w_m / (k + \text{rank}_m(d))$ .                        |
| VQA Exact Match            | Answer / VQA                 | Percentage of questions for which the normalized generated answer exactly matches the reference answer.                                                                       |
| VQA faithfulness           | Answer / VQA                 | Percentage of answers supported by the resolved media evidence and the requested visual content.                                                                              |
| Hallucination rate         | Answer / VQA                 | Fraction of answers unsupported by valid media evidence or contradicted by the evidence. A fail-closed refusal is not counted as a hallucination.                             |
| Ambiguity score            | Query / KIS-C                | Combined ambiguity signal from Distinct Video Ratio (DVR) and Score Margin Ambiguity (SMA). Lower values indicate stronger convergence after clarification.                   |
| Ambiguity reduction        | Query / KIS-C                | Difference between the initial and final ambiguity scores: $A_{\text{initial}} - A_{\text{final}}$ . Higher reduction indicates more effective conversational disambiguation. |
| Latency (p50/p95)          | System / operation           | Median or 95th-percentile wall-clock time for the complete measured operation. We report seconds for VLM workflows and milliseconds for retrieval paths.                      |
| Throughput (QPS)           | System / concurrency         | Completed query or candidate-scoring operations per second under the stated worker count.                                                                                     |
| Recall versus exact        | Retrieval / precision ladder | Recall of an approximate HNSW mode relative to the exact brute-force ranking over the same collection.                                                                        |
| Overall RAG score          | Benchmark / corpus           | Aggregate benchmark score combining retrieval quality, grounded answer quality, conversational behavior, and operational measurements under the declared evaluation protocol. |